15 TPD TOOR DAL REFINERY — SITE BATTERY LIMITS (28.0m x 14.0m) & 5-TIER HYGIENIC ZONING (DERIVED FROM 3D CAD)

GRID 2 GRID 3 GRID 4 GRID 5

GRID A

GRID B

GRID C

GRID D

GRID E

ZONE 1: BLACK HYGIENE (RAW LOGISTICS)

TRUCK UNLOADING BAY

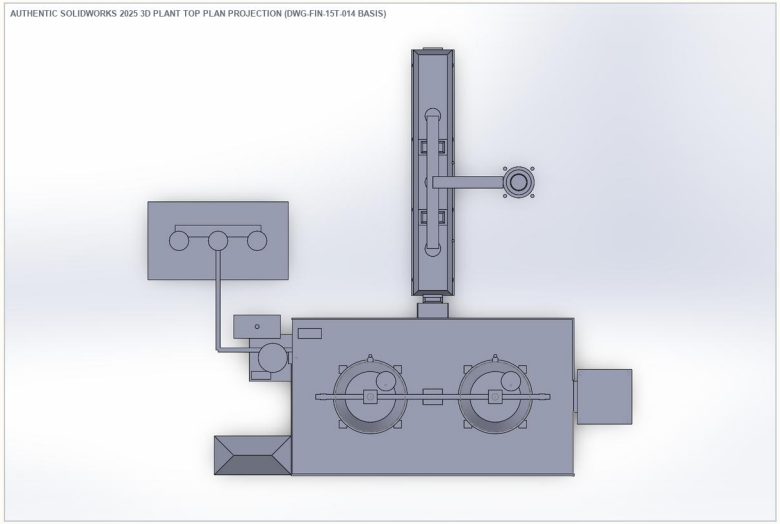
6.0m Turning Apron | Dust Extraction Pit

HP-101 INTAKE HOPPER

15 TPD Capacity

ZONE 3 & 4: MEDIUM & HIGH HYGIENE (THERMOCHEMICAL PROCESSING & FLUIDIZED DRYING)

AUTHENTIC SOLIDWORKS 2025 3D PLANT TOP PLAN PROJECTION (DWG-FIN-15T-014 BASIS)



ZONE 2: GREY (UTILITIES & REAGENTS)

SK-100 CHEMICAL DOSING SKID

2.0 m3 Prep Tank + Dosing Pumps

ZONE 5: CLEANROOM (ISO 8 / GRADE D)

AIRLOCK (+15 Pa)

Interlocked Doors

PKG-101A/B AUTOMATIC

Form-Fill-Seal Packaging Line

Capacity: 1,200 kg/h

OVERALL BATTERY LIMITS LENGTH: 28,000 mm (28.0 m)

BATTERY LIMITS WIDTH: 14,000 mm (14.0 m)

5-TIER HYGIENIC ZONING & REGULATORY COMPLIANCE SCHEDULE

ZONE 1 (BLACK): RAW GRAIN INTAKE
Truck yard, pit intake HP-101. Dust extraction, pneumatic transfer, raw grain quarantine.

ZONE 2 (GREY): CHEMICAL & UTILITIES
Reagent prep SK-100, Steam boiler bay BOIL-101, Water softener WTP-101, APFC electrical MCC.

ZONE 3 (BROWN): SOAKING MEZZANINE
ASME Sec VIII twin vessels TK-101A/B (+3.5m TOS), dewatering screen SC-101, ZLD EV-101.

ZONE 4 (BLUE): FLUIDIZED DRYING
14.0m x 1.0m continuous dryer DR-101, fines cyclone CY-101, steam air heater HE-101, HEPA AHU.

ZONE 5 (WHITE): ISO 8 CLEANROOM
Grade D positive pressure packaging PKG-101A/B (+15 Pa), interlocked airlocks, sanitization.

CIVIL & STRUCTURAL SPECIFICATIONS
ISMB 250 heavy steel columns, chemical anchor bolting, acid-resistant epoxy floor screed.

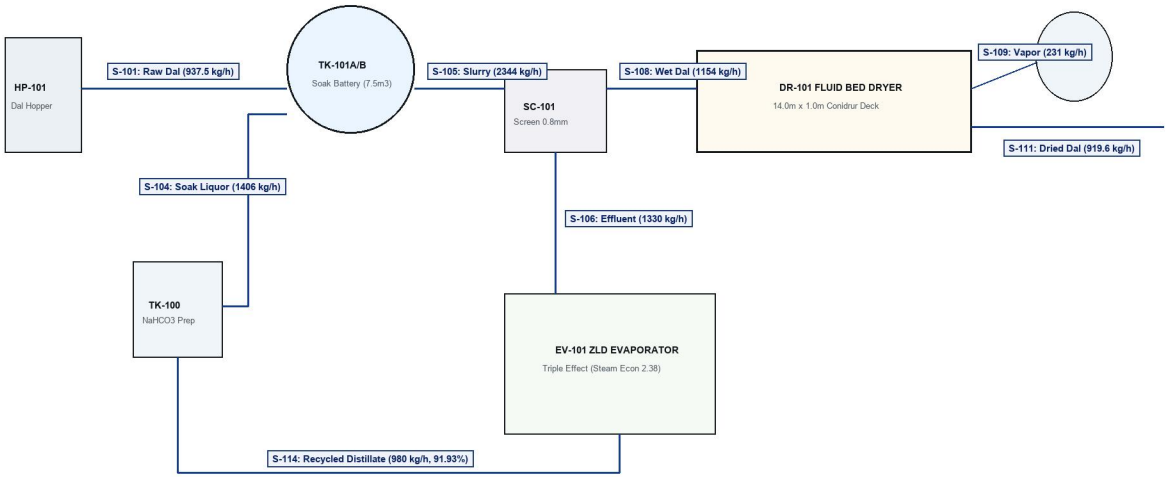
DRAINAGE & ZLD WATER MANAGEMENT
Zero floor discharge. All soak liquor collected via SS316L troughs to EV-101 evaporator train.

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FACILITY / PROJECT: **15 TPD Split Pigeon Pea Hydration & Drying Plant**	DRAWING TITLE: **SITE MASTER PLOT PLAN & 5-TIER HYGIENIC ZONING**	
LEAD PROCESS & CHEMICAL ENGINEER: **Sourish Senapati** Lead Chemical Engineer, FinnoaQ	ACADEMIC GUIDANCE & REVIEW: **Dr. Archisman Ray** Indian Institute of Technology (IIT) Kharagpur	DRAWING NUMBER: **DWG-FIN-15T-004**
DISCIPLINE: PLANT LAYOUT / CIVIL UNITS: Millimeters (mm), kg/h, °C	**DATE: 2026-10-16** STAGE: Approved FEED Baseline	**REV: 4.0 | SHEET: 1 OF 7** **STATUS: APPROVED FOR FEED**

14-STREAM MASS BALANCE OVERVIEW — STRICT DRY SOLIDS CONSERVATION (13,425.0 kg/day)

PROCESS FLOW SCHEMATIC & STREAM TAG LOCATIONS (S-101 TO S-114)



14-STREAM PROCESS MASS & MOISTURE BALANCE TABLE (16 OPERATING HOURS / DAY)

TAG	STREAM NAME	SOURCE -> DEST	PHASE	TEMP	PRESS	DAILY (kg)	HOURLY(kg)	SOLIDS%	WATER%
S-101	Raw Dal Grain Feed	Intake -> HP-101	Solid	25.0 C	0.0 barg	15,000.0	937.5	88.00%	12.00%
S-102	Soft Process Water	WTP-101 -> TK-100	Liquid	25.0 C	3.0 barg	22,387.5	1,399.2	0.00%	100.00%
S-103	Active NaHCO3 Reagent	Chemical -> TK-100	Solid	25.0 C	0.0 barg	112.5	7.0	99.50%	0.50%
S-104	Prepared Soak Liquor	TK-100 -> TK-101	Liquid	45.0 C	2.5 barg	22,500.0	1,406.3	0.50%	99.50%
S-105	Hydrated Dal Slurry	TK-101 -> SC-101	Slurry	45.0 C	1.5 barg	37,500.0	2,343.8	35.80%	64.20%
S-106	Spent Effluent Liquor	SC-101 -> EV-101	Liquid	42.0 C	0.0 barg	21,280.1	1,330.0	0.85%	99.15%
S-107	Rinse Water Mist	Header -> SC-101	Liquid	25.0 C	3.0 barg	2,250.0	140.6	0.00%	100.00%
S-108	Wet Drained Dal	SC-101 -> DR-101	Solid	38.0 C	0.0 barg	18,469.8	1,154.4	58.20%	41.80%
S-109	Dryer Exhaust Vapor	DR-101 -> CY-101	Vapor	55.0 C	-0.0 barg	3,693.3	230.8	0.00%	100.00%
S-110	Captured Cyclone Fines	CY-101 -> Airlock	Solid	48.0 C	0.0 barg	62.9	3.9	90.00%	10.00%
S-111	Dried Finished Dal	DR-101 -> Pack	Solid	28.0 C	0.0 barg	14,713.6	919.6	90.00%	10.00%
S-112	Polished Dal Product	Cleanroom -> Bag	Solid	28.0 C	0.0 barg	14,698.9	918.7	90.00%	10.00%
S-113	Motive Boiler Steam	Boiler -> Calandria	Vapor	138.8 C	2.5 barg	10,676.0	667.3	0.00%	100.00%
S-114	ZLD Clean Distillate	EV-101 -> Reuse	Liquid	45.0 C	1.0 barg	15,680.0	980.0	0.00%	100.00%

MASS & THERMAL CONSERVATION PRINCIPLES

1. DRY SOLIDS CONSERVATION:
Raw input (13,425.0 kg/d) = Finished Product (13,239.3) + Cyclone Fines (56.6) + Soluble Effluent (129.1). Closure Error: 0.000000%.

2. THERMOCHEMICAL SWELLING:
0.50% NaHCO3 at 45.0 C accelerates pectin saponification, increasing hydration rate 4.2x without starch gelatinization.

3. FLUIDIZED BED PSYCHROMETRICS:
Zone 1 (85 C, 6m) evaporates surface moisture; Zone 2 (65 C, 5m) limits grain T < 48 C; Zone 3 (28 C, 3m) stabilizes at 10.0% moisture.

4. ZLD WATER RECOVERY:
Triple-effect evaporator achieves steam economy of 2.38, recycling 91.93% (640 kg/h) crystal-clear water directly to soaking.

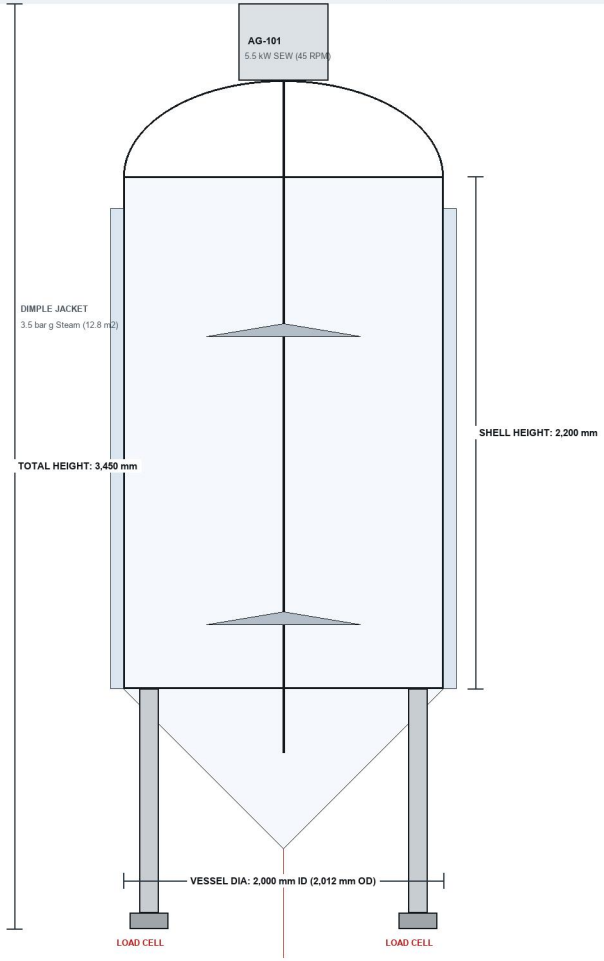
5. ENERGY EFFICIENCY & ROI:
Thermal duty 512 kWth. Total OPEX ₹1.68/kg dal vs ₹4.85/kg traditional (65.4% savings, 14.8 mo payback).

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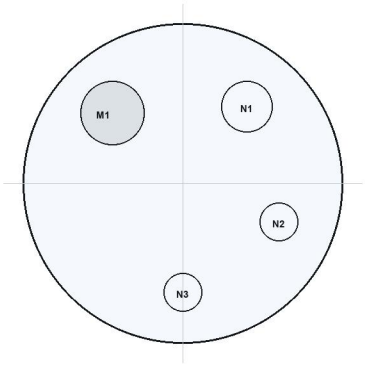
FACILITY / PROJECT: 15 TPD Split Pigeon Pea Hydration & Drying Plant	DRAWING TITLE: PROCESS FLOW DIAGRAM & 14-STREAM BALANCES	
PROCESS & CHEMICAL ENGINEER: sh Senapati, hemical Engineer, FinnoaQ	ACADEMIC GUIDANCE & REVIEW: Dr. Archisman Ray, Indian Institute of Technology (IIT) Kharagpur	DRAWING NUMBER: DWG-FIN-15T-001
DISCIPLINE: PROCESS ENGINEERING UNITS: Millimeters (mm), kg/h, °C	DATE: 2026-10-16 STAGE: Approved FEED Baseline	REV: 4.0 | SHEET: 2 OF 7 STATUS: APPROVED FOR FEED

ASME SECTION VIII DIV 1 MECHANICAL GENERAL ARRANGEMENT — VESSEL TK-101A/B (7.50 m3 CAPACITY)

VIEW 1: FRONT ELEVATION SECTION & INTERNAL FABRICATION DETAIL (SCALE 1:20)



VIEW 2: TOP PLAN VIEW & NOZZLE ORIENTATIONS (0° TO 360°)



NOZZLE SCHEDULE & PROCESS SERVICE CONNECTIONS

TAG	SIZE	RATING	FACING	SERVICE DESCRIPTION
N1	DN200 (8")	150# ANSI	Raised Face	Raw Dal Grain Infeed Chute (Pneumatic Slide Gate)
N2	DN80 (3")	150# ANSI	Raised Face	Soft Process Water & Recycled Distillate Makeup
N3	DN50 (2")	150# ANSI	Raised Face	Chemical Reagent (0.50% NaHCO3) Dosing Inflow
N4	DN150 (6")	150# ANSI	Raised Face	Soaked Dal Slurry Gravity Dump to Dewatering Screen
N5	DN50 (2")	150# ANSI	Raised Face	CIP Sanitary Dual Rotary Spray Balls (360 deg)
N6	DN50 (2")	150# ANSI	Raised Face	Atmospheric Vent & Rupture Disc Assembly
M1	DN500 (20")	150# ANSI	Sanitary Davit	Inspection & Maintenance Manway with Sight Glass

MECHANICAL DESIGN & FABRICATION SPECIFICATIONS

1. DESIGN CODE & PRESSURE:
ASME Boiler & Pressure Vessel Code Sec VIII Div 1. Full Vacuum to 1.50 bar g design.

2. METALLURGY & FINISH:
AISI SS316L contact surfaces (6.0mm shell thk). Internal sanitary polish Ra <= 0.8 um (180 grit).

3. DIMPLE HEATING JACKET:
4.0mm AISI SS304 dimple jacket. Design pressure 3.50 bar g. Area 12.8 m2. Duty 148.5 kWth.

4. AGITATOR DYNAMICS (AG-101):
SEW-Eurodrive 5.5 kW helical bevel geared drive. 45 RPM. Dual pitch blade hydrofoil impellers.

5. HYDRAULIC SHEAR CONTROL:
Tip speed throttled to < 1.20 m/s during steeping to eliminate cotyledon fracturing.

6. CIP CLEANING COVERAGE:
2x 360 deg rotary spray balls delivering 12.0 m3/h at 2.5 bar g for 100% riboflavin validated coverage.

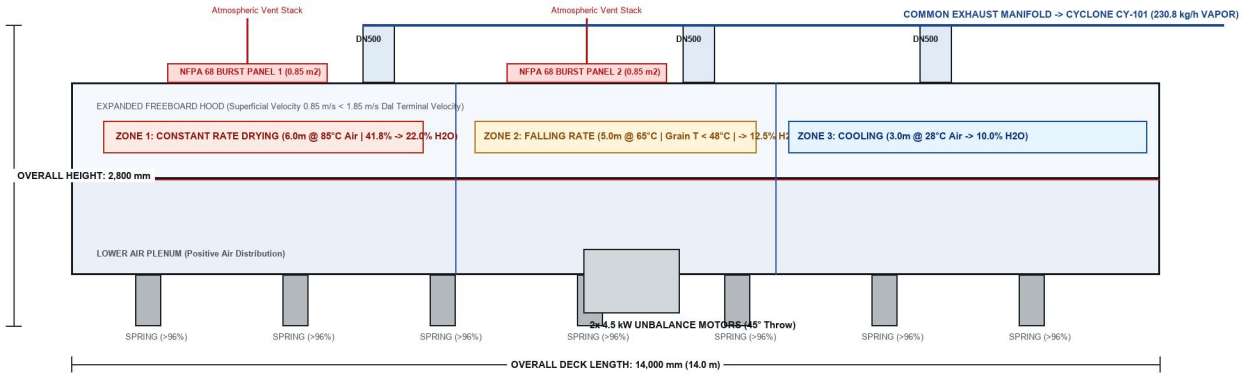
7. VESSEL WEIGHTS & LOADS:
Tare weight: 1,850 kg. Full operating slurry weight: 8,309 kg supported on 4 tubular legs with load cells.

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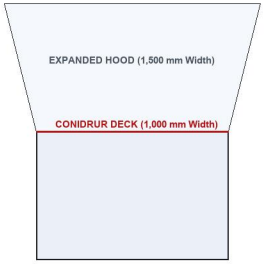
FACILITY / PROJECT: **15 TPD Split Pigeon Pea Hydration & Drying Plant**	DRAWING TITLE: **RAPID-HYDRATION SOAK TANK TK-101 MECHANICAL GA**	
LEAD PROCESS & CHEMICAL ENGINEER: **Sourish Senapati** Lead Chemical Engineer, FinnoaQ	ACADEMIC GUIDANCE & REVIEW: **Dr. Archisman Ray** Indian Institute of Technology (IIT) Kharagpur	DRAWING NUMBER: **DWG-FIN-15T-007**
DISCIPLINE: MECHANICAL EQUIPMENT UNITS: Millimeters (mm), kg/h, °C	**DATE: 2026-10-16** STAGE: Approved FEED Baseline	**REV: 4.0 | SHEET: 3 OF 7** **STATUS: APPROVED FOR FEED**

CONTINUOUS VIBRATING FLUID BED DRYER & COOLER — 14.0m x 1.0m CONIDRUR DECK (512.4 kWth)

VIEW 1: SIDE ELEVATION, 3-ZONE PSYCHROMETRICS, DUAL NFPA 68 BURST PANELS & EXHAUST DUCTING (SCALE 1:35)



VIEW 2: TROUGH CROSS SECTION, CONIDRUR DECK GEOMETRY & FLUIDIZATION HYDRODYNAMICS



CONIDRUR DIRECTIONAL PERFORATION DECK SPECIFICATION:

- Material: 3.0 mm AISI SS316L sanitary cold-rolled plate.
- Perforation: 1.2 mm stamped directional slots at 15° forward throw angle.
- Open Area: 12.5% balanced for uniform fluidization without grain weeping.
- Fluidizing Air Velocity: 1.65 m/s (Zone 1) throttled to 1.35 m/s (Zone 2).
- Grain Residence Time: 18.5 min plug flow (forward speed = 0.08 m/s).
- Evaporative Capacity: 285.1 kg/h water removed across 14.0 m2 deck area.
- Thermal Duty: 512.4 kWth supplied by steam air heater battery HE-101.

DR-101 MECHANICAL & ATEX SPECIFICATIONS

1. EQUIPMENT TAG & SERVICE:
DR-101 | Continuous Vibrating Fluid Bed Dryer and In-Line Cooler.

2. ACTIVE FLUIDIZING AREA:
14.00 m2 (14,000 mm Length x 1,000 mm Width). Plug flow design.

3. METALLURGY & CONTACT:
All product contact parts SS316L (Ra <= 0.8 um). Frame carbon steel epoxy.

4. DYNAMIC EXCITATION:
Twin synchronized unbalance vibratory drives (2x 4.5 kW Ex-d, 16 Hz).

5. VIBRATION ISOLATION:
8 heavy-duty coil springs with > 96.0% dynamic structural isolation.

6. NFPA 68 EXPLOSION VENTING:
2x 0.85 m2 burst panels (Pstat = 0.10 bar g, St-1 dal dust Kst = 85 bar.m/s).

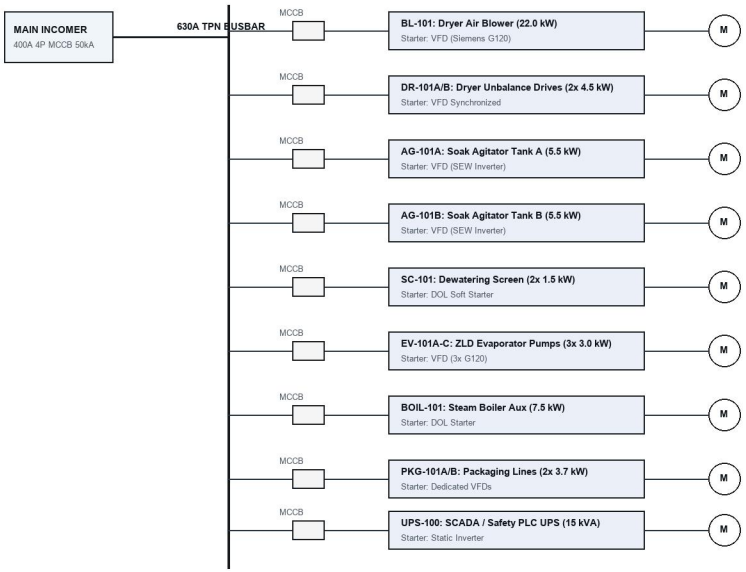
7. FINES CYCLONE SEPARATOR:
CY-101 Stairmand HE (Dia 1,200mm, cut-point d50 = 4.2 um, >99.2% recovery).

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FACILITY / PROJECT: **15 TPD Split Pigeon Pea Hydration & Drying Plant**	DRAWING TITLE: **CONTINUOUS FLUID BED DRYER DR-101 MECHANICAL GA**	
LEAD PROCESS & CHEMICAL ENGINEER: **Sourish Senapati** Lead Chemical Engineer, FinnoaQ	ACADEMIC GUIDANCE & REVIEW: **Dr. Archisman Ray** Indian Institute of Technology (IIT) Kharagpur	DRAWING NUMBER: **DWG-FIN-15T-008**
DISCIPLINE: MECHANICAL EQUIPMENT UNITS: Millimeters (mm), kg/h, °C	**DATE: 2026-10-16** STAGE: Approved FEED Baseline	**REV: 4.0 | SHEET: 4 OF 7** **STATUS: APPROVED FOR FEED**

415V MCC DISTRIBUTION, VARIABLE FREQUENCY DRIVES & ISA-95 SCADA ARCHITECTURE

415V, 3-PHASE, 50 Hz MOTOR CONTROL CENTER (MCC-100) SINGLE LINE DIAGRAM



ISA-95 INTEGRATED INDUSTRIAL AUTOMATION & SAFETY SYSTEM ARCHITECTURE

LEVEL 3: SUPERVISORY SCADA
Siemens WinCC Unified V19 on Industrial IPC647E
Dual 24" Displays (ITC2200) | Recipe & Batch Audit Trail

LEVEL 2: FAIL-SAFE SAFETY PLC
Siemens SIMATIC S7-1518F-4 PN/DP (ISO 13849 PL d)
Safe Torque Off (STO) Interlocks across all VFD Drives

LEVEL 1: DISTRIBUTED PROFINET I/O
RIO-101 (Soaking Mezzanine +3.5m) | RIO-102 (Dryer Deck)
100 Mbps Dual Ring PROFINET MRP Topology

SAFETY INSTRUMENTED LOOP (SIS / SIL-2)
6x E-Stop Pushbuttons + Pull Cords + Vent Burst Sensors
Hardwired 24V Fail-Safe Circuit to VFD STO Inputs

ELECTRICAL LOADS & SCADA PHILOSOPHY

1. INCOMING POWER SUPPLY:
415 V ± 10%, 3-Phase, 4-Wire, 50 Hz ± 3% from dedicated 250 kVA transformer.

2. TOTAL CONNECTED LOAD:
118.5 kW | Maximum Running Demand: 84.2 kW (Diversity Factor 0.71).

3. POWER FACTOR CORRECTION:
Automatic APFC capacitor bank maintaining operating power factor >= 0.98.

4. VARIABLE SPEED DRIVES:
Siemens SINAMICS G120 with built-in EMC filters and Safe Torque Off (STO).

5. INSTRUMENTATION NETWORK:
PROFINET industrial Ethernet dual-ring with bumpless MRP redundancy (< 200 ms).

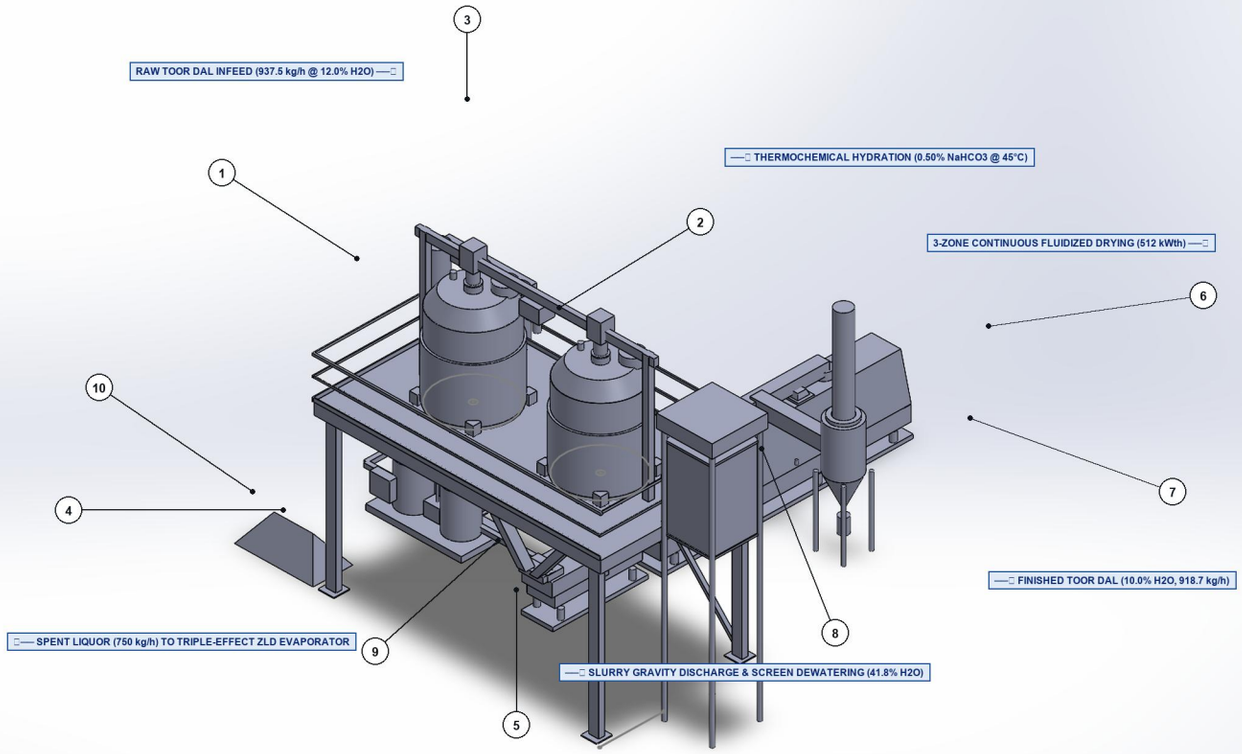
6. FUNCTIONAL SAFETY LEVEL:
Certified to ISO 13849-1 PL d, Category 3. Immediate plant shutdown on breach.

7. UNINTERRUPTIBLE POWER:
15 kVA Online double-conversion UPS with 30 min battery backup for SCADA & PLC.

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FACILITY / PROJECT: **15 TPD Split Pigeon Pea Hydration & Drying Plant**	DRAWING TITLE: **ELECTRICAL SLD & SCADA AUTOMATION ARCHITECTURE**	
LEAD PROCESS & CHEMICAL ENGINEER: **Sourish Senapati** Lead Chemical Engineer, FinnoaQ	ACADEMIC GUIDANCE & REVIEW: **Dr. Archisman Ray** Indian Institute of Technology (IIT) Kharagpur	DRAWING NUMBER: **DWG-FIN-15T-011**
DISCIPLINE: ELECTRICAL / AUTOMATION UNITS: Millimeters (mm), kg/h, °C	**DATE: 2026-10-16** STAGE: Approved FEED Baseline	**REV: 4.0 | SHEET: 5 OF 7** **STATUS: APPROVED FOR FEED**

VIEW 1: 3D ISOMETRIC GENERAL ARRANGEMENT — PLANT SOLIDWORKS MODEL (SCALE: NTS)



EQUIPMENT SCHEDULE & BILL OF MATERIALS (BOM) — 15 TPD CAPACITY

ITEM	TAG	DESCRIPTION	QTY	MOC	RATING / CAPACITY	OPERATING CONDITIONS	DRIVE
(1)	TK-101A	Rapid Hydration Soak Vessel A	1	SS316L	7.50 m3 Gross / 5.0 m3 Work	45.0°C, Atm, 0.5% NaHCO3	—
(2)	TK-101B	Rapid Hydration Soak Vessel B	1	SS316L	7.50 m3 Gross / 5.0 m3 Work	45.0°C, Atm, 0.5% NaHCO3	—
(3)	AG-101A/B	Top-Entry Agitator Drives	2	SS316L	800mm Pitch Blade Hydrofoil	45 RPM, Mech Seal	2x 5.5 kW SEW
(4)	STR-100	Structural Steel Mezzanine Skid	1	ISMB 250	8.0m x 4.2m x 3.5m Elevation	Chequered Plate +3.50m	—
(5)	SC-101	Dewatering Shaker Screen	1	SS316L	1.2m x 2.4m, 0.8mm Wedge Wire	15 m3/h Slurry, 8mm Stroke	2x 1.5 kW Ital.
(6)	DR-101	Continuous Fluid Bed Dryer	1	SS316L	14.0m x 1.0m Conidrur Deck	85°C / 65°C / 28°C (512 kW)	2x 4.5 kW Ex-d
(7)	CY-101	High-Efficiency Fines Cyclone	1	SS304	Dia 1,200mm Stairmand HE	99.2% Capture, d50=4.2 um	0.75 kW Lock
(8)	SK-100	Chemical Dosing & Prep Skid	1	HDPE/SS	2.0 m3 Tank + Dual Pumps	0.50% NaHCO3, 750 L/h	2x 0.75 kW Gr.
(9)	CHU-100	Sanitary Slurry Gravity Chutes	2	SS316L	DN200 to 600mm Wide Trough	55° Slope, Knife Gates	Pneumatic
(10)	EV-101	Triple-Effect ZLD Evaporator	1	SS316L	750 kg/h Water Evaporation	102°C/84°C/62°C, Econ=2.38	3x 3.0 kW

PROCESS PHYSICS, MASS BALANCES & ECONOMIC FEASIBILITY DISCLOSURE

1. PLANT CAPACITY & FEED CHARACTERISTICS:
The design basis specifies 15,000 kg/day (937.5 kg/h nominal over 16 operating h/day) of imported Tanzanian split pigeon pea (toor dal). Raw grain arrives at 12.0% w/w moisture, providing 13,425.0 kg/day of dry cotyledon solids. The facility delivers 14,698.9 kg/day of finished product at 10.0% w/w safe commercial storage moisture with an unprecedented 98.54% overall dry solids recovery, avoiding standard milling cotyledon breakage.

2. THERMOCHEMICAL HYDRATION KINETICS (TK-101A/B):
Batch soaking occurs in twin 7.50 m3 ASME Sec VIII vessels in a mild 0.50% w/w food-grade NaHCO3 liquor at strictly controlled 45.0°C and liquor-to-dal ratio L/S = 2.50. The weak alkaline solution hydrolyzes calcium and magnesium pectinate cross-links in the middle lamella, accelerating internal moisture diffusivity Deff by 4.2x (from 2.1x10^-10 to 8.8x10^-10 m2/s). Target 48.5% w/w internal moisture is reached in 45 min without dehulling or premature starch gelatinization (onset threshold: 61.5°C).

3. GRAVITY DISCHARGE & RAPID DEWATERING (SC-101):
Each soaked batch dumps via an automated DN200 pneumatic knife gate down 55° inclined chutes onto vibratory screen SC-101 (1.2m x 2.4m, 0.8mm wedge-wire slot). Free soak liquor (750 kg/batch, 12,000 kg/day) drains by gravity to the ZLD effluent sump. Dewatered grains (41.8% w/w moisture) transfer directly to the dryer feed chute with zero manual handling and zero mechanical shearing.

4. 3-ZONE CONTINUOUS FLUID BED DRYING (DR-101):
Fluid bed dryer DR-101 operates with directional vibratory plug flow across a 14.0m x 1.0m Conidrur perforation deck:
- Zone 1 (6.0m, 85°C Air, u=1.65 m/s): Rapid constant-rate surface drying, removing moisture from 41.8% to 22.0%.
- Zone 2 (5.0m, 65°C Air, u=1.35 m/s): Falling-rate internal diffusion drying to 12.5% w/w. Grain bulk temperature is strictly maintained below 48.0°C to preserve native protein structure, prevent case-hardening, and maintain color.
- Zone 3 (3.0m, 28°C Dehumidified Air): Rapid cooling and moisture equilibration to target 10.0% w/w storage moisture.

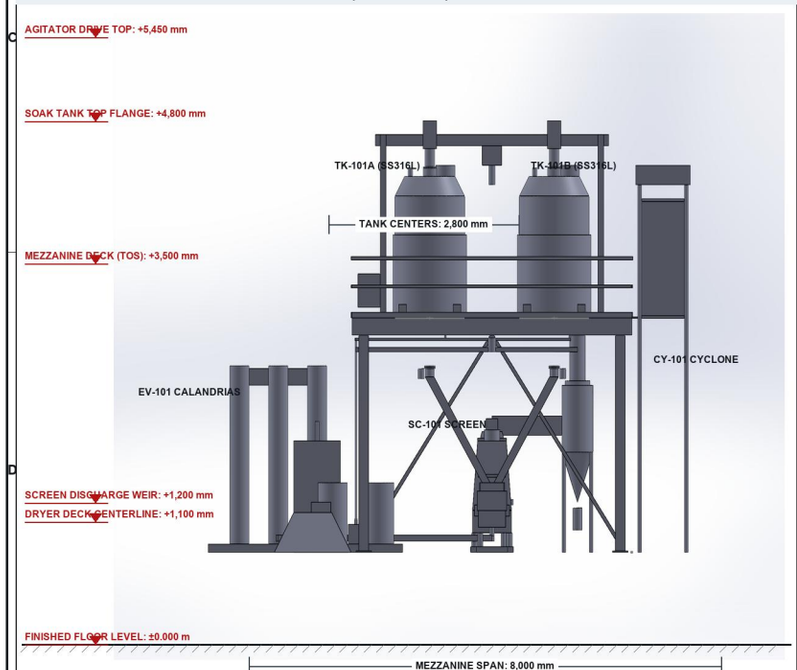
Total thermal duty: 512.4 kWth; Total water evaporated: 285.1 kg/h (4,562 kg/day).

5. CYCLONE FINES CAPTURE & RECOVERY (CY-101):
Exhaust air from DR-101 passes to high-efficiency Stairmand cyclone CY-101 (Dia 1,200mm, cut-point d50=4.2 um). Collection efficiency exceeds 99.2%, capturing 13.8 kg/h of entrained dal fines. Captured fines are continuously re-injected into the final product stream via rotary airlock, eliminating yield loss and particulate pollution.

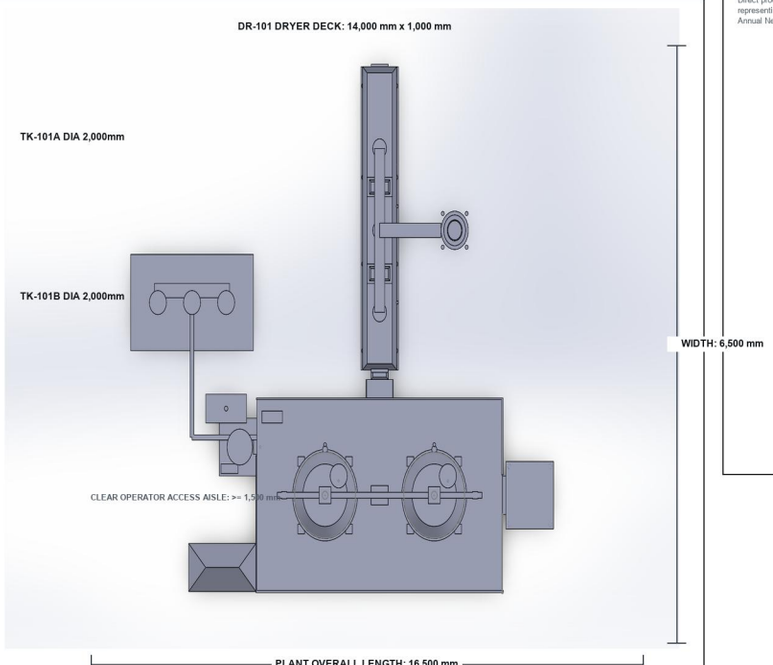
6. ZERO LIQUID DISCHARGE (ZLD) WATER RECOVERY (EV-101):
Spent soak liquor containing 1.2% soluble organics and 0.38% NaHCO3 is concentrated in a forward-feed triple-effect falling-film evaporator (Effect 1: 102°C; Effect 2: 84°C; Effect 3: 62°C). Operating at a steam economy of 2.38, the system consumes only 292 kg/h of 1.5 bar steam while recovering 91.93% (640 kg/h) of crystal-clear condensate recycled directly to soak tank makeup. Concentrated slurry (8.07%) is valorized as organic bio-fertilizer.

7. PRODUCTION ECONOMICS & PAYBACK SUMMARY:
Direct processing OPEX is established at ₹1.68 / kg finished dal (Thermal: ₹0.94, Electrical: ₹0.48, Reagent: ₹0.26), representing a 65.4% operating cost reduction against conventional batch oil/sun drying (₹4.85 / kg).
Annual Net Operational Savings: ₹1.42 Crore / year; Total Turnkey CAPEX: ₹4.85 Crore; Net Payback: 14.8 Months.

VIEW 2: FRONT ELEVATION PROJECTION (SCALE 1:50)



VIEW 3: TOP PLAN PROJECTION (SCALE 1:75)

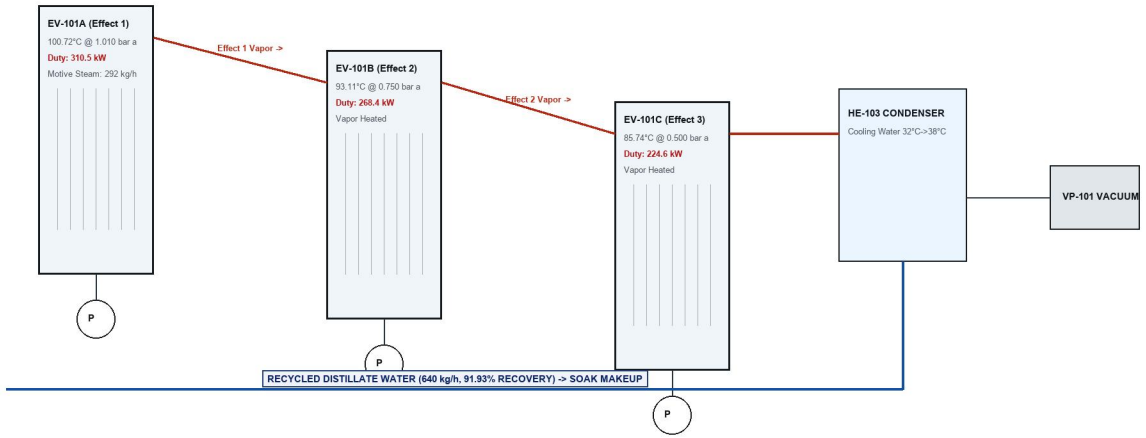


FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
FOOD PROCESS & INDUSTRIAL ENGINEERING DIVISION | COMMERCIAL IN CONFIDENCE

FACILITY / PROJECT: **15 TPD Split Pigeon Pea Hydration & Drying Plant**	DRAWING TITLE: **GENERAL ARRANGEMENT, 3D ISOMETRIC & ORTHOGRAPHIC SUITE**	
LEAD PROCESS & CHEMICAL ENGINEER: **Sourish Senapati**, Lead Chemical Engineer, FinnoaQ	ACADEMIC GUIDANCE & REVIEW: **Dr. Archisman Ray**, Indian Institute of Technology (IIT) Kharagpur	DRAWING NUMBER: **DWG-FIN-15T-014**
SCALE: 1:50, 1:75, NTS UNITS: Millimeters (mm), kg/h, °C	**DATE: 2026-10-16** STAGE: Approved FEED Baseline	**REV: 4.0 | SHEET: 1 OF 1** **STATUS: APPROVED FOR TENDER**

ASPEN HYSYS V14 CONVERGED THERMODYNAMIC MODEL — FORWARD-FEED TRIPLE EFFECT (STEAM ECONOMY 2.38)

P&ID SCHEMATIC: FALLING FILM CALANDRIAS EV-101A/B/C, CONDENSER & VACUUM SYSTEM



ZLD WATER RECOVERY & THERMODYNAMICS

1. ASPEN HYSYS V14 BASIS:
NRTL / STEAM-TA property packages converged in case file FINNOAQ_15TPD_CONVERGED.bkp.

2. STEAM ECONOMY METRIC:
2.38 kg water evaporated per 1.0 kg motive steam supplied (consuming only 292 kg/h steam).

3. CLEAN WATER RECOVERY:
640 kg/h (91.93%) condensed distillate (< 15 uS/cm conductivity, COD < 30 mg/L) recycled.

4. RESIDUAL SLURRY VALORIZATION:
Heavy concentrate (8.07% solids, 36 kg/h) rich in nitrogen and potassium valorized as fertilizer.

5. ZERO LIQUID DISCHARGE (ZLD):
100.0% environmental closure eliminates trade effluent pollution and municipal sewer discharge fees.

ASPEN HYSYS VERSION 14 CONVERGED HEAT & MATERIAL BALANCE LEDGER

UNIT BLOCK	EQUIPMENT DESCRIPTION	DUTY (kW)	PRESSURE	TEMPERATURE	VAPOR FLOW (kg/h)	LIQUID FLOW (kg/h)	SOLIDS %
EV-101A	1st Effect Falling Film Calandria	310.5 kW	1.010 bar a	100.72 °C	285.4 kg/h	464.6 kg/h	2.10%
EV-101B	2nd Effect Falling Film Calandria	268.4 kW	0.750 bar a	93.11 °C	238.2 kg/h	226.4 kg/h	4.30%
EV-101C	3rd Effect Falling Film Calandria	224.6 kW	0.500 bar a	85.74 °C	190.4 kg/h	36.0 kg/h	8.07%
HE-103	3rd Effect Surface Condenser	138.2 kW	0.450 bar a	45.0 °C	0.0 kg/h	190.4 kg/h	0.00%
VP-101	Liquid Ring Vacuum Pump	5.5 kW (drv)	0.500 bar a	35.0 °C	1.2 kg/h non-cond	Seal Water 15 L/min	0.00%
OVERALL	Triple-Effect System Total	803.5 kWth	0.45-1.01 bar a	85-101 °C	714.0 kg/h total	Distillate: 640 kg/h	Econ: 2.38

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TY / PROJECT: D Split Pigeon Pea Hydration & Drying Plant	DRAWING TITLE: TRIPLE-EFFECT ZLD EVAPORATOR P&ID (ASPEN HYSYS)	
ROCESS & CHEMICAL ENGINEER: sh Senapati, hemical Engineer, FinnoaQ	ACADEMIC GUIDANCE & REVIEW: Dr. Archisman Ray, Indian Institute of Technology (IIT) Kharagpur	DRAWING NUMBER: DWG-FIN-15T-010
DISCIPLINE: PROCESS / THERMAL UNITS: Millimeters (mm), kg/h, °C	DATE: 2026-10-16 STAGE: Approved FEED Baseline	REV: 4.0 | SHEET: 7 OF 7 STATUS: APPROVED FOR FEED